```
            var expiry =
Instant.ofEpochSecond(json.getInt("exp"));    #C
            var username = json.getString("sub");    #C
            var attrs = json.getJSONObject("attrs");    #C

            var token = new Token(expiry, username);    #C
            for (var key : attrs.keySet()) {    #C
                token.attributes.put(key,
attrs.getString(key));    #C
            }

            return Optional.of(token);
        } catch (JSONException e) {
            return Optional.empty();
        }
    }

    @Override
    public void revoke(Request request, String tokenId) {
        // TODO    #D
    }
}
```

#A Convert the token attributes into a JSON object.
#B Encode the JSON object with URL-safe base64-encoding.
#C To read the token, decode it and parse the JSON to recover the attributes.
#D Leave the revoke method blank for now.

6.1.1 Protecting JSON tokens with HMAC

Of course, as it stands this code is completely insecure. Anybody can log in to the API and then edit the encoded